Statement of Work
Appendix B

1. Equipment Details/Photos
2. Observation Report from ENG Firm
3. Medical Gas Certification Report 2026

System	Manufacturer	Model #	Serial #	Location / Room	Details
Medical Vacuum	Dekker	VMX0203KA3-40-ES	I19576	1st Floor Mech. Room C106	• Approx. capacity: 73 scfm @ 19"HG per pump • Triplex (3 pumps) • oil-sealed liquid ring • separator reservoir • receiver tank

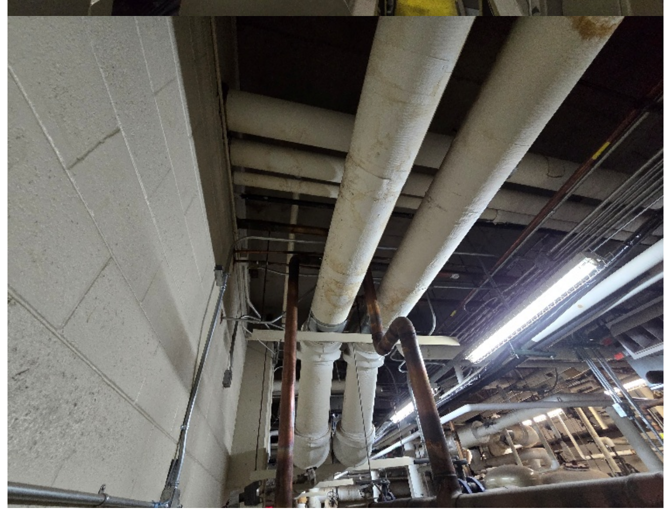
CARDINA
ENGINEERING • COMMISSIONING
(915) 373-6030 10600 Montwood Dr., Ste 124
www.cardinagroup.com El Paso, TX 79935
Texas Engineering Firm No. F-20011

Site Observation Report

Project:	El Paso VA HCS- Vacuum System Equipment Analysis
Subject:	Site Observation
Date:	Friday, January 02, 2026
Location:	El Paso, Texas
Attendees:	Aaron Medina (CARDINA), Carlos Arguijo (CARDINA), Joe Hurtado (VA)

No.	Observation	Photo
1.	CARDINA Engineering Commissioning conducted a site observation at the VA Hospital located at 5001 N Piedras St, El Paso, TX 79930. The purpose of visit was to do a visual assessment of the vacuum system equipment.	
2.	The vacuum system is a Dekker oil-sealed liquid ring vacuum system, air-cooled, with a separator tank. The system has three pumps, each with a nominal capacity of 200 cfm at atmospheric pressure. The model no. is VMX0203KA3-40-ES The nameplate information indicates the system was manufactured in November of 2007.	
3.	Reservoir tank has “U” stamp.	

No.	Observation	Photo
4.	The runtime on the pumps is as follows: Pump 1: 37,545 hours Pump 2: 41,871 hours Pump 3: 31,378 hours Two of the pumps are used approximately 25% more than Pump 3. It is unknown if the lead/lag sequence is not operating properly or a pump was down for extended period of time.	
5.	View of pumps and controllers, isolation valves present at each pump..	
6.	A non-uniform gap was observed at one of the system discharge vent flanges.	

No.	Observation	Photo
		
7.	The existing piping does not have hangers or supports at every change of direction. It is suggested these be added.	 

No.	Observation	Photo
8.	Temperatures at motor cooling fins: 153 deg. F. Temperatures at fan discharge: 146 deg. F.	 
9.	Maintenance logs or formal records were not observed on site. A handwritten note indicated that the motor was greased on 1/30/2012; however, no additional documentation was available to verify subsequent maintenance activities. At the time of the site visit, staff were unable to provide maintenance records or confirm established maintenance procedures for the equipment.	

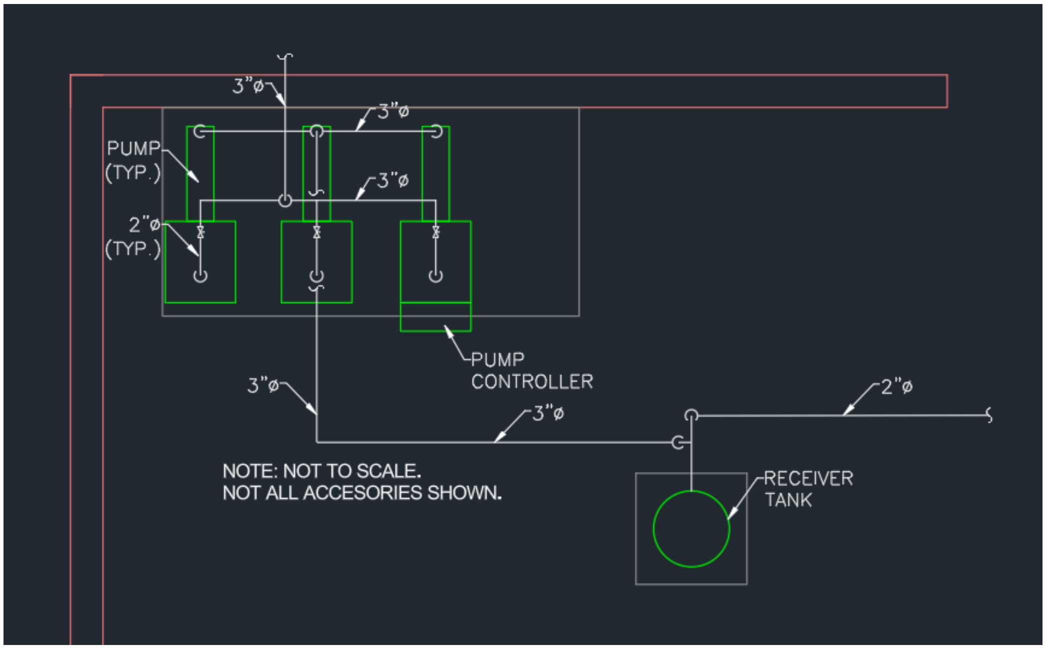


Figure 1: Sketch of vacuum pump and receiver layout.

Summary:

The existing installation appears to generally comply with NFPA 99 requirements. However, review of pump usage data and the absence of documented maintenance suggest that the periodic maintenance required by NFPA 99 may not have been consistently performed. All three pumps have accumulated over 30,000 operating hours. Considering the extended service life and the uncertainty regarding maintenance, it is recommended that the vacuum system be replaced.

Respectfully,

Carlos E. Arguijo

MEDICAL GAS ANNUAL INSPECTION

2026

MGI™ INC.
MEDICAL GAS INSTALLERS

EL PASO VAMC
5001 N PIEDRAS ST
EL PASO, TX 79930

PREPARED BY: BRAYDEN SEGER
SERVICE DATE: JANUARY 6TH, 2026

MGI INC.
MEDICAL GAS INSTALLERS

MEDICAL GAS ANNUAL TESTING ANALYSIS

ITEMS TESTED/INSPECTED	MEDICAL GAS TYPE							
	OXYGEN	MEDICAL AIR	MEDICAL VACUUM	NITROUS OXIDE	NITROGEN	CARBON DIOXIDE	INST. AIR	WAGD
SOURCE EQUIPMENT	✓	✓	✓	✓	✓			
MASTER ALARMS	✓	✓	✓	✓	✓			
ZONE VALVES	✓	✓	✓	✓	✓			✓
ZONE VERIFICATION	✓	✓	✓	✓	✓			✓
CROSS CONNECTION	✓	✓	✓	✓	✓			✓
AREA ALARMS	✓	✓	✓	✓	✓			
PATIENT OUTLETS	✓	✓	✓	✓	✓			✓
PARTICLE ANALYSIS	✓	✓		✓	✓			
PURITY ANALYSIS	✓	✓		✓	✓			

SCOPE OF TESTING/VERIFICATION	NFPA 99 2018 Compliant Medical Gas Annual Inspection for a Category 1 Medical Facility. Includes testing of all medical gas source equipment, medical gas zone valves, medical gas master alarm panels, medical gas area alarm panels, medical gas control panels, medical gas flexible terminals and medical gas terminals and medical gas labeling.
ADDITIONAL COMMENTS	

MGI MEDICAL GAS VERIFIER: BRAYDEN SEGER

SIGNATURE: Brayden Seger

DATE: 1/6/2026

CLICK OR SCAN TO VIEW NITC CARD

EXECUTIVE SUMMARY

MGI™ INC.

MEDICAL GAS INSTALLERS

EXECUTIVE OVERALL SUMMARY-SOURCE

SOURCE EQUIPMENT				
GAS TYPE	SOURCE TYPE	TOTAL COUNT	TOTAL PASSED	TOTAL FAILED
OXYGEN	BULK OXYGEN	1	1	0
MEDICAL AIR	RECIPROCATING	1	1	0
VACUUM	LIQUID RING	1	1	0
NITROUS OXIDE	GAS X GAS MANIFOLD	1	1	0
NITROGEN	GAS X GAS MANIFOLD	1	1	0
DENTAL AIR	SCROLL	1	1	0
DENTAL VACUUM	TURBINE	1	1	0
MASTER ALARM PANEL	ALARMS	2	2	0
TOTAL		9	9	0

SOURCE EQUIPMENT MASTER ALARMS					
EQUIPMENT	EQUIPMENT MAKE & MODEL	ALARM PANEL COUNT	ALARM SIGNAL COUNT	SIGNALS PASSED	SIGNALS FAILED
MASTER ALARM PANEL	OHIO HEALTHCAIR	1	23	19	4
MASTER ALARM PANEL	OHIO HEALTHCAIR	1	23	20	3
TOTALS		2	46	39	7

MEDICAL GAS INSTALLERS

EXECUTIVE OVERALL SUMMARY-PIPELINE

ZONE VALVE BOXES		
TOTAL ZVB LOCATIONS	TOTAL PASSED	
13	10	
MISSING VALVES	FAILED	REPAIRED
0	3	0

AREA ALARM PANELS		
TOTAL AREA ALARM PANELS	TOTAL PASSED	
5	5	
MISSING ALARMS	FAILED	REPAIRED
0	0	0

PATIENT OUTLET TERMINALS				
GAS TYPE	TOTAL COUNT	TOTAL PASSED	TOTAL FAILED	TOTAL UNAVAILABLE
OXYGEN	78	78	0	0
MEDICAL AIR	53	53	0	0
VACUUM	104	104	0	0
NITROUS OXIDE	16	15	1	0
NITROGEN	25	25	0	0
WAGD	16	16	0	0
TOTAL	292	291	1	0
PERCENTAGE		99.7%	0.3%	0.0%

TOTAL PERCENTAGE OF OUTLETS TESTED	100.0%

TOTAL OUTLET FAILURES	TOTAL OUTLETS REPAIRED
1	0

OUTLET ISSUES

EL PASO VAMC

MGI TECHNICIAN: BRAYDEN SEGER

1	ROOM	SPECIFIC OUTLET	OUTLET SPECS	ISSUE	REPAIRED BY	REPAIR DATE
	OR-6 LEFT COLUMN	NITROUS OXIDE	ALLIED - CNT2 - CH	LEAKS W/OUT ADAPTER INSERTED		

END OF OUTLET ISSUES

CORRECTIVE ACTIONS

MGI INC. MEDICAL GAS INSTALLERS

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA	ISSUE	NFPA 99 2018 CODE	CORRECTED
ZONE VALVE FOR MEDICAL AIR SERVING BIOMED	NORMAL OPERATING PRESSURE DOES NOT FALL IN THE MIDDLE THIRD OF THE ZONE VALVE GAUGE.	***NFPA 99 2018 CODE 5.1.8.1.3**** *The scale range of positive pressure analog indicators shall be such that the normal operating pressure is within the middle third of the total range [e.g., an indicator of 0 to 2070 kPa (0 to 300 psi) would have a lower third of 0 to 690 kPa (0 to 100 psig), a middle third of 690 kPa to 1380 kPa (100 psig to 200 psig), and a top third of 1380 kPa to 2070 kPa (200 psig to 300 psig)].*	
CORRECTIVE ACTION	INTALL A 0-100 PSI GAUGE ON THE MEDICAL AIR ZONE VALVE		

AREA	ISSUE	NFPA 99 2018 CODE	CORRECTED
ZONE VALVE FOR MEDICAL AIR SERVING DENTAL CLINIC	NORMAL OPERATING PRESSURE DOES NOT FALL IN THE MIDDLE THIRD OF THE ZONE VALVE GAUGE.	***NFPA 99 2018 CODE 5.1.8.1.3**** *The scale range of positive pressure analog indicators shall be such that the normal operating pressure is within the middle third of the total range [e.g., an indicator of 0 to 2070 kPa (0 to 300 psi) would have a lower third of 0 to 690 kPa (0 to 100 psig), a middle third of 690 kPa to 1380 kPa (100 psig to 200 psig), and a top third of 1380 kPa to 2070 kPa (200 psig to 300 psig)].*	
CORRECTIVE ACTION	INTALL A 0-100 PSI GAUGE ON THE MEDICAL AIR ZONE VALVE		

AREA	ISSUE	NFPA 99 2018 CODE	CORRECTED
ZONE VALVE FOR NITROUS OXIDE SERVING OR-4	NORMAL OPERATING PRESSURE DOES NOT FALL IN THE MIDDLE THIRD OF THE ZONE VALVE GAUGE.	***NFPA 99 2018 CODE 5.1.8.1.3**** *The scale range of positive pressure analog indicators shall be such that the normal operating pressure is within the middle third of the total range [e.g., an indicator of 0 to 2070 kPa (0 to 300 psi) would have a lower third of 0 to 690 kPa (0 to 100 psig), a middle third of 690 kPa to 1380 kPa (100 psig to 200 psig), and a top third of 1380 kPa to 2070 kPa (200 psig to 300 psig)].*	
CORRECTIVE ACTION	INSTALL 0-100 PSI GAUGE ON THE NITROUS OXIDE ZONE VALVE		

AREA	ISSUE	NFPA 99 2018 CODE	CORRECTED
PRESSURE SWITCH/GAUGES FOR OXYGEN, VACUUM, NITROGEN, AND NITROUS OXIDE	PRESSURE SWITCHES/GAUGES ARE NOT ON GAS SPECIFIC DISS DEMAND CHECKS.	***NFPA 99 2018 CODE 5.1.8.2.3*** *All pressure-sensing devices and main line pressure gauges downstream of the source valves shall be provided with a gas-specific demand check fitting to facilitate service testing or replacement.*	
CORRECTIVE ACTION	INSTALL GAS SPECIFIC DISS DEMAND CHECKS ON THE PRESSURE SWITCH/GAUGE.		

MGI INC.
MEDICAL GAS INSTALLERS

CORRECTIVE ACTIONS

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA	ISSUE	NFPA 99 2018 CODE	CORRECTED
MASTER ALARMS	OXYGEN LINE PRESSURE HIGH DOES NOT HAVE A SIGNAL RAN TO THE MASTER ALARM PANELS	***Master alarm panels for medical gas and vacuum systems shall each include the following signals: (NFPA 99 2018 5.1.9.2.4) (7) Alarm indication when the pressure in the main line of each separate medical gas system increases 20 percent or decreases 20 percent from the normal operating pressure.***	
CORRECTIVE ACTION	WIRE IN THE OXYGEN LINE PRESSURE HIGH TO BOTH MASTER ALARM PANELS.		

AREA	ISSUE	NFPA 99 2018 CODE	CORRECTED
MASTER ALARMS	THE DENTAL AIR HIGH LINE PRESSURE ALARMED AS LOW LINE PRESSURE AND THE LOW LINE PRESSURE ALARMED AS HIGH LINE PRESSURE	***Master alarm panels for medical gas and vacuum systems shall each include the following signals: (NFPA 99 2018 5.1.9.2.4) (7) Alarm indication when the pressure in the main line of each separate medical gas system increases 20 percent or decreases 20 percent from the normal operating pressure.***	
CORRECTIVE ACTION	REWIRE THE PRESSURE SWITCH TO FIX THE MISMATCHED ALARMS.		

AREA	ISSUE	NFPA 99 2018 CODE	CORRECTED
MASTER ALARMS	THE MEDICAL AIR HIGH DEWPOINT DID NOT ALARM AT THE ENGINEERING BUILDING MASTER ALARM	***Master alarm panels for medical gas and vacuum systems shall each include the following signals: (NFPA 99 2018 5.1.9.2.4) (10) Medical air dew point high alarm from each compressor site to indicate when the line pressure dew point is greater than +2°C (+35°F)***	
CORRECTIVE ACTION	CHECK WIRING AT THE MASTER ALARM PANEL AND THE PRESSURE SWITCH.		

NOTE: CORRECTIVE ACTIONS ARE COMPONENTS OF THE MEDICAL GAS SYSTEM THAT DO NOT MEET THE NFPA 99 2018 CODE AND SHOULD BE UPGRADED OR REPAIRED TO MEET THE NFPA 99 2018 CODE.

END OF CORRECTIVE ACTIONS

FUTURE CORRECTIONS

MEDICAL GAS INSTALLERS

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA	ISSUE	NFPA 99 2018 CODE
NONE		
FUTURE CORRECTION		

NOTE: THESE ARE COMPONENTS OF THE MEDICAL GAS SYSTEM THAT DO NOT MEET THE 2018 NFPA 99 CODE, HOWEVER THEY ARE PERMITTED TO REMAIN IN USE UNTIL THE AREA(S) ARE UPGRADED OR RENOVATED DUE TO THE ABOVE NOT AFFECTING PATIENT CARE.

END OF FUTURE CORRECTIONS

MEDICAL GAS MASTER ALARM ANALYSIS

MGI™ INC.

MEDICAL GAS INSTALLERS

MASTER ALARM CODE REQUIREMENTS

Master alarm panels for medical gas and vacuum systems shall each include the following signals: (NFPA 99 2018 5.1.9.2.4)

(1) Alarm indication when, or just before, changeover occurs in a medical gas system that is supplied by a manifold or other alternating-type bulk system that has as a part of its normal operation a changeover from one portion of the operating supply to another.

(2) Alarm indication for a bulk cryogenic liquid system when the main supply reaches an average day's supply, indicating low contents.

(3) Alarm indication when, or just before, the changeover to the reserve supply occurs in a medical gas system that consists of one or more units that continuously supply the piping system while another unit remains as the reserve supply and operates only in the case of an emergency.

(4) Alarm indication for cylinder reserve pressure low when the content of a cylinder reserve header is reduced below one average day's supply.

(5) For bulk cryogenic liquid systems, an alarm when or at a predetermined set point before the reserve supply contents fall to one day's average supply, indicating reserve low.

(6) Where a cryogenic liquid storage unit is used as a reserve for a bulk supply system, an alarm indication when the gas pressure available in the reserve unit is below that required for the medical gas system to function.

(7) Alarm indication when the pressure in the main line of each separate medical gas system increases 20 percent or decreases 20 percent from the normal operating pressure.

(8) Alarm indication when the medical–surgical vacuum pressure in the main line of each vacuum system drops.

(9) Alarm indication(s) from the local alarm panel(s) as described in 5.1.9.5.2 to indicate when one or more of the conditions being monitored at a site is in alarm.

(10) Medical air dew point high alarm from each compressor site to indicate when the line pressure dew point is greater than +2°C (+35°F)

(11) WAGD low alarm when the WAGD vacuum level or flow is below effective operating limits.

(12) An instrument air dew point high alarm from each compressor site to indicate when the line pressure dew point is greater than −30°C (−22°F).

(13) Alarm indication if the primary or reserve production stops on a proportioning system.

MASTER ALARM ANALYSIS

EL PASO VAMC
EL PASO, TX 79930

MGI TECHNICIAN: BRAYDEN SEGER
TESTING DATE: 1/6/2026

MASTER ALARM SIGNALS	VISUAL STATUS	AUDIBLE STATUS	LABELED CORRECTLY	PASS/FAIL	COMMENTS
OXYGEN PRIMARY LIQUID LEVEL LOW	✓	✓	✓	PASS	
OXYGEN RESERVE IN USE	✓	✓	✓	PASS	
OXYGEN RESERVE LOW	✓	✓	✓	PASS	
OXYGEN LINE PRESSURE HIGH				FAIL	NO SIGNAL RAN FOR HIGH
OXYGEN LINE PRESSURE LOW	✓	✓	✓	PASS	
MEDICAL AIR RESERVE COMP IN USE	✓	✓	✓	PASS	
MEDICAL AIR HIGH DEW POINT	×	×	✓	FAIL	DID NOT ALARM AT PANEL
MEDICAL AIR GENERAL ALARM	✓	✓	✓	PASS	
MEDICAL AIR LINE PRESSURE HIGH	✓	✓	✓	PASS	
MEDICAL AIR LINE PRESSURE LOW	✓	✓	✓	PASS	
MEDICAL VACUUM LAG IN USE	✓	✓	✓	PASS	
MEDICAL VACUUM LINE PRESSURE LOW	✓	✓	✓	PASS	
MEDICAL VACUUM HIGH TEMP SHUTDOWN	✓	✓	✓	PASS	
NITROGEN RESERVE IN USE	✓	✓	✓	PASS	
NITROGEN LINE PRESSURE HIGH	✓	✓	✓	PASS	
NITROGEN LINE PRESSURE LOW	✓	✓	✓	PASS	
NITROUS OXIDE RESERVE IN USE	✓	✓	✓	PASS	
NITROUS OXIDE LINE PRESSURE HIGH	✓	✓	✓	PASS	
NITROUS OXIDE LINE PRESSURE LOW	✓	✓	✓	PASS	
DENTAL AIR SYSTEM GENERAL ALARM	✓	✓	✓	PASS	
DENTAL AIR HIGH DEW POINT	✓	✓	✓	PASS	
DENTAL AIR LOW LINE PRESSURE	✓	✓	×	FAIL	ALARMED AS HIGH LINE PRESSURE
DENTAL AIR HIGH LINE PRESSURE	✓	✓	×	FAIL	ALARMED AS LOW LINE PRESSURE
ALARM MAKE AND MODEL:	OHIO HEALTHCAIR			**LOCATION:**	ENGINEERING BUILDING

PER NFPA 99 2018 WHERE IS THE MASTER ALARM LOCATED OF THE FOLLOWING LOCATIONS	LOCATION	COMMENTS
(1) One master alarm panel shall be located in the office or work space of the on-site individual responsible for the maintenance of the medical gas and vacuum piping systems. (2) In order to ensure continuous surveillance of the medical gas and vacuum systems while the facility is in operation, the second master alarm panel shall be located in an area of continuous observation (e.g., the telephone switchboard, security office, or other continuously staffed location).	1 **PASS/FAIL** FAIL	

MEDICAL GAS INSTALLERS

MASTER ALARM ANALYSIS

EL PASO VAMC
EL PASO, TX 79930

MGI TECHNICIAN: BRAYDEN SEGER
TESTING DATE: 1/6/2026

MASTER ALARM SIGNALS	VISUAL STATUS	AUDIBLE STATUS	LABELED CORRECTLY	PASS/FAIL	COMMENTS
OXYGEN PRIMARY LIQUID LEVEL LOW	✓	✓	✓	PASS	
OXYGEN RESERVE IN USE	✓	✓	✓	PASS	
OXYGEN RESERVE LOW	✓	✓	✓	PASS	
OXYGEN LINE PRESSURE HIGH				FAIL	NO SIGNAL RAN FOR HIGH
OXYGEN LINE PRESSURE LOW	✓	✓	✓	PASS	
MEDICAL AIR RESERVE COMP IN USE	✓	✓	✓	PASS	
MEDICAL AIR HIGH DEW POINT	✓	✓	✓	PASS	
MEDICAL AIR GENERAL ALARM	✓	✓	✓	PASS	
MEDICAL AIR LINE PRESSURE HIGH	✓	✓	✓	PASS	
MEDICAL AIR LINE PRESSURE LOW	✓	✓	✓	PASS	
MEDICAL VACUUM LAG IN USE	✓	✓	✓	PASS	
MEDICAL VACUUM LINE PRESSURE LOW	✓	✓	✓	PASS	
MEDICAL VACUUM HIGH TEMP SHUTDOWN	✓	✓	✓	PASS	
NITROGEN RESERVE IN USE	✓	✓	✓	PASS	
NITROGEN LINE PRESSURE HIGH	✓	✓	✓	PASS	
NITROGEN LINE PRESSURE LOW	✓	✓	✓	PASS	
NITROUS OXIDE RESERVE IN USE	✓	✓	✓	PASS	
NITROUS OXIDE LINE PRESSURE HIGH	✓	✓	✓	PASS	
NITROUS OXIDE LINE PRESSURE LOW	✓	✓	✓	PASS	
DENTAL AIR SYSTEM GENERAL ALARM	✓	✓	✓	PASS	
DENTAL AIR HIGH DEW POINT	✓	✓	✓	PASS	
DENTAL AIR LOW LINE PRESSURE	✓	✓	×	FAIL	ALARMED AS HIGH LINE PRESSURE
DENTAL AIR HIGH LINE PRESSURE	✓	✓	×	FAIL	ALARMED AS LOW LINE PRESSURE
ALARM MAKE AND MODEL:	OHIO HEALTHCAIR			LOCATION:	BIOMED

PER NFPA 99 2018 WHERE IS THE MASTER ALARM LOCATED OF THE FOLLOWING LOCATIONS	LOCATION	COMMENTS
(1) One master alarm panel shall be located in the office or work space of the on-site individual responsible for the maintenance of the medical gas and vacuum piping systems. (2) In order to ensure continuous surveillance of the medical gas and vacuum systems while the facility is in operation, the second master alarm panel shall be located in an area of continuous observation (e.g., the telephone switchboard, security office, or other continuously staffed location).	2 PASS/FAIL FAIL	

MEDICAL GAS AREA ALARM ANALYSIS

MGI™ INC.

MEDICAL GAS INSTALLERS

AREA ALARM CODE REQUIREMENTS

Area alarm systems used for medical gas and vacuum systems shall include the following: (NFPA 99 2018 5.1.9.1)

(1) Separate visual indicators for each condition monitored, except as permitted in 5.1.9.5.2 for local alarms that are displayed on master alarm panels.

(2) Visual indicators that remain in alarm until the situation that has caused the alarm is resolved.

(3) Cancelable audible indication of each alarm condition that produces a sound with a minimum level of 80 dBA at 0.92m (3 ft)

(4) Means to visually indicate a lamp or LED failure

(5) Visual and audible indication that the communication with an alarm-initiating devices is disconnected.

(6) Labeling of each indicator, indication the condition monitored.

(7) Labeling of each alarm panel for its area of surveillance.

(8) Reinitiating of the audible signal if another alarm condition occurs while the audible alarm is silenced.

(9) Power for master, area alarms, sensors, and switches from the life safety branch of the emergency electrical system as described in Chapter 6.

Area Alarms. Area alarm panels shall be provided to monitor all medical gas, medical–surgical vacuum, and piped WAGD systems supplying the following: (NFPA 99 2018 5.1.9.4)

(1) Anesthetizing locations where moderate sedation, deep sedation, or general anesthesia is administered

(2) *Critical Care Areas

Area alarm panels for medical gas systems shall indicate if the pressure in the lines in the area being monitored increases or decreases by 20 percent from the normal line pressure. (NFPA 99 2018 5.1.9.4.2)

Area alarm panels for medical–surgical vacuum systems shall indicate if the vacuum in the area drops to or below 300 mm (12 in.) gauge HgV. (NFPA 99 2018 5.1.9.4.3)

Alarm sensors for area alarms shall be located as follows: (NFPA 99 2018 5.1.9.4.4)

(1)*Critical care areas shall have the alarm sensors installed on the patient or use side of each individual zone valve box assemblies.

(2)*Anesthetizing locations where moderate sedation, deep sedation, or general anesthesia is administered shall have the sensors installed either on the source side of any of the individual room zone valve box assemblies or on the patient or use side of each of the individual zone valve box assemblies.

MGI™ INC.
MEDICAL GAS INSTALLERS

AREA ALARM ANALYSIS
3RD FLOOR

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA MONITORED:	GAS MONITORED	PRESSURE READING	LABELING CORRECT	HIGH ALARM SETPOINT	LOW ALARM SETPOINT	VISUAL ALARM FUNCTION	AUDIBLE ALARM FUNCTION	MAINLINE OR ZONE SENSOR	PASS	FAIL	COMMENTS
RADIOLOGY ROOMS B309, B312, B315, B317	OXYGEN	53 PSI	PASS	60 PSI	40 PSI	PASS	PASS	ZONE	✓		
	MED AIR	46 PSI	PASS	60 PSI	40 PSI	PASS	PASS	ZONE	✓		
	VACUUM	23" HG	PASS	N/A	12" HG	PASS	PASS	ZONE	✓		
FLOOR: 3RD	**ALARM LOCATION:** RADIOLOGY CHECK IN					**ALARM MAKE AND MODEL:** CHEMETRON IMPACT					

END OF 3RD FLOOR AREA ALARM ANALYSIS

AREA ALARM ANALYSIS
4TH FLOOR

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA MONITORED:	GAS MONITORED	PRESSURE READING	LABELING CORRECT	HIGH ALARM SETPOINT	LOW ALARM SETPOINT	VISUAL ALARM FUNCTION	AUDIBLE ALARM FUNCTION	MAINLINE OR ZONE SENSOR	PASS	FAIL	COMMENTS
ASU RECOVERY BAYS 1-16	OXYGEN	56 PSI	PASS	60 PSI	40 PSI	PASS	PASS	ZONE	✓		
	MED AIR	47 PSI	PASS	60 PSI	40 PSI	PASS	PASS	ZONE	✓		
	VACUUM	21" HG	PASS	N/A	12" HG	PASS	PASS	ZONE	✓		
FLOOR: 4TH	ALARM LOCATION: NURSES STATION					ALARM MAKE AND MODEL: CHEMETRON IMPACT					

AREA MONITORED:	GAS MONITORED	PRESSURE READING	LABELING CORRECT	HIGH ALARM SETPOINT	LOW ALARM SETPOINT	VISUAL ALARM FUNCTION	AUDIBLE ALARM FUNCTION	MAINLINE OR ZONE SENSOR	PASS	FAIL	COMMENTS
PROCEDURE ROOMS B459 AND B457	OXYGEN	55 PSI	PASS	60 PSI	40 PSI	PASS	PASS	ZONE	✓		
	VACUUM	21" HG	PASS	N/A	12" HG	PASS	PASS	ZONE	✓		
FLOOR: 4TH	ALARM LOCATION: ACROSS CHECK IN DESK					ALARM MAKE AND MODEL: CHEMETRON IMPACT					

AREA MONITORED:	GAS MONITORED	PRESSURE READING	LABELING CORRECT	HIGH ALARM SETPOINT	LOW ALARM SETPOINT	VISUAL ALARM FUNCTION	AUDIBLE ALARM FUNCTION	MAINLINE OR ZONE SENSOR	PASS	FAIL	COMMENTS
DENTAL CLINIC	OXYGEN	54 PSI	PASS	60 PSI	40 PSI	PASS	PASS	ZONE	✓		
	MED AIR	48 PSI	PASS	60 PSI	40 PSI	PASS	PASS	ZONE	✓		
	VACUUM	21" HG	PASS	N/A	12" HG	PASS	PASS	ZONE	✓		
FLOOR: 4TH	ALARM LOCATION: OUTSIDE B421					ALARM MAKE AND MODEL: CHEMETRON IMPACT					

AREA MONITORED:	GAS MONITORED	PRESSURE READING	LABELING CORRECT	HIGH ALARM SETPOINT	LOW ALARM SETPOINT	VISUAL ALARM FUNCTION	AUDIBLE ALARM FUNCTION	MAINLINE OR ZONE SENSOR	PASS	FAIL	COMMENTS
OR'S 1-8	OXYGEN	55 PSI	PASS	60 PSI	40 PSI	PASS	PASS	MAINLINE	✓		
	MED AIR	46 PSI	PASS	60 PSI	40 PSI	PASS	PASS	MAINLINE	✓		
	VACUUM	20" HG	PASS	N/A	12" HG	PASS	PASS	MAINLINE	✓		
	N2O	52 PSI	PASS	60 PSI	40 PSI	PASS	PASS	MAINLINE	✓		
	NITROGEN	160 PSI	PASS	195 PSI	155 PSI	PASS	PASS	MAINLINE	✓		
FLOOR: 4TH	ALARM LOCATION: CLEAN CORE					ALARM MAKE AND MODEL: CHEMETRON IMPACT					

END OF 4TH FLOOR AREA ALARM ANALYSIS

MEDICAL GAS
ZONE VALVE ANALYSIS

MGI INC.™

MEDICAL GAS INSTALLERS

MEDICAL GAS INSTALLERS

ZONE VALVE CODE REQUIREMENTS

Zone Valves. All station outlets/inlets shall be supplied through a zone valve as follows:
(NFPA 99 2018 5.1.4.6.1)

(1) The zone valve shall be placed such that a wall intervenes between the valve and outlets/inlets that it controls.

(2) The zone valves shall serve only outlets/inlets located on that same story.

(3) The zone valve shall not be located in a room with station outlets/inlets that it controls.

Zone valves shall be readily operable from a standing position in the corridor on the same floor they serve. (NFPA 99 2018 5.1.4.6.2)

Zone valves shall be so arranged that shutting off the supply of medical gas or vacuum to one zone will not affect the supply of medical gas or vacuum to another zone or the rest of the system.
(NFPA 99 2018 5.1.4.6.3)

A pressure/vacuum indicator shall be provided on the station outlet/inlet side of each zone valve.
(NFPA 99 2018 5.1.4.6.4)

Zone valve boxes shall be installed where they are visible and accessible at all times.
(NFPA 99 2018 5.1.4.6.1)

Zone valve boxes shall not be installed behind normally open or normally closed doors or otherwise hidden from plain view. (NFPA 99 2018 5.1.4.6.1)

Zone valve boxes shall not be located in closed or locked rooms, areas, or closets.
(NFPA 99 2018 5.1.4.6.1)

Zone valves shall be so arranged that shutting off the supply of gas to any one operating room or anesthetizing location will not affect the others. (NFPA 99 2018 5.1.4.6.3)

Zone valves shall be labeled in accordance with 5.1.11.2.7

MEDICAL GAS INSTALLERS

ZONE VALVE ANALYSIS
1ST FLOOR

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
BIOMED	OXYGEN	58 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		MED AIR GAUGE NEEDS TO BE O-100 PSI
	MED AIR	50 PSI	PASS	PASS	1/2" BALL	PASS	PASS		✕	
	VACUUM	22" HG	PASS	PASS	1" BALL	PASS	PASS	✓		
	N2	160 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
FLOOR: 1ST	**VALVE LOCATION:** OUTSIDE BIOMED				**VALVE MANUFACTURER:** CHEMETRON					

END OF 1ST FLOOR ZONE VALVE ANALYSIS

ZONE VALVE ANALYSIS
3RD FLOOR

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
RADIOLOGY ROOMS B309, B312, B315, B317	OXYGEN	56 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	MED AIR	52 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	VACUUM	22" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
FLOOR: 3RD	**VALVE LOCATION:** ACROSS B317				**VALVE MANUFACTURER:** CHEMETRON					

END OF 3RD FLOOR ZONE VALVE ANALYSIS

MGI INC. MEDICAL GAS INSTALLERS

ZONE VALVE ANALYSIS
4TH FLOOR

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
ASU RECOVERY BAYS 1-16	OXYGEN	59 PSI	PASS	PASS	1" BALL	PASS	PASS	✓		
	MED AIR	52 PSI	PASS	PASS	1" BALL	PASS	PASS	✓		
	VACUUM	25" HG	PASS	PASS	1-1/2" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** OUTSIDE RECOVERY				**VALVE MANUFACTURER:** CHEMETRON					

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
PROCEDURE ROOMS B459 AND B457	OXYGEN	58 PSI	PASS	PASS	3/4" BALL	PASS	PASS	✓		
	VACUUM	25" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** OUTSIDE PROCEDURE ROOMS				**VALVE MANUFACTURER:** CHEMETRON					

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
DENTAL CLINIC	OXYGEN	53 PSI	PASS	PASS	3/4" BALL	PASS	PASS	✓		MEDICAL AIR NEEDS A 0-100 PSI GAUGE
	MED AIR	50 PSI	PASS	PASS	3/4" BALL	PASS	PASS		✗	
	VACUUM	24" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** ACROSS B414				**VALVE MANUFACTURER:** CHEMETRON					

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
OR-1 (A-405)	OXYGEN	58 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	MED AIR	50 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	VACUUM WAGD	24" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
	N2O	51 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	N2	170 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** OUTSIDE OR-1				**VALVE MANUFACTURER:** CHEMETRON					

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
OR-2 (A-406)	OXYGEN	58 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	MED AIR	58 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	VACUUM WAGD	21" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
	N2O	50 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	N2	165 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** OUTSIDE OR-2				**VALVE MANUFACTURER:** CHEMETRON					

MGI INC. MEDICAL GAS INSTALLERS

ZONE VALVE ANALYSIS
4TH FLOOR

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
OR-3 (B-473)	OXYGEN	58 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	MED AIR	50 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	VACUUM WAGD	22" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
	N2O	55 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	N2	170 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** OUTSIDE OR-3				**VALVE MANUFACTURER:** CHEMETRON					

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
OR-4 (B-474)	OXYGEN	55 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		N2O NEEDS A 0-100 PSI GAUGE
	MED AIR	48 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	VACUUM	24" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
	N2O	55 PSI	PASS	PASS	1/2" BALL	PASS	PASS		✗	
	N2	160 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** OUTSIDE OR-4				**VALVE MANUFACTURER:** CHEMETRON					

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
OR-5 (B-475)	OXYGEN	59 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	MED AIR	48 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	VACUUM	22" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
	N2O	50 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	N2	175 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** ACROSS OR-5				**VALVE MANUFACTURER:** CHEMETRON					

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
OR-6 (D-403)	OXYGEN	58 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	MED AIR	50 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	VACUUM	26" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
	N2O	53 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	N2	175 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** OUTSIDE OR-6				**VALVE MANUFACTURER:** CHEMETRON					

MEDICAL GAS INSTALLERS

ZONE VALVE ANALYSIS
4TH FLOOR

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
OR-7 (C-409)	OXYGEN	56 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	MED AIR	48 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	VACUUM	23" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
	N2O	54 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	N2	165 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** OUTSIDE OR-7				**VALVE MANUFACTURER:** CHEMETRON					

AREA SERVED:	GAS	GAUGE READING	GAS LABELED CORRECTLY	ZONE LABELED CORRECTLY	VALVE SIZE & TYPE	PROPER LOCATION	LEAKAGE	PASS	FAIL	COMMENTS
OR-8 (C-408)	OXYGEN	59 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	MED AIR	50 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	VACUUM	26" HG	PASS	PASS	3/4" BALL	PASS	PASS	✓		
	N2O	52 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
	N2	170 PSI	PASS	PASS	1/2" BALL	PASS	PASS	✓		
FLOOR: 4TH	**VALVE LOCATION:** OUTSIDE OR-8				**VALVE MANUFACTURER:** CHEMETRON					

END OF 4TH FLOOR ZONE VALVE ANALYSIS

MEDICAL GAS
OUTLET ANALYSIS

MGI™ INC.

MEDICAL GAS INSTALLERS

MEDICAL GAS INSTALLERS

OUTLET CODE REQUIREMENTS

Each station outlet/inlet for medical gases or vacuums shall be gas-specific, whether the outlet/inlet is threaded or is a noninterchangeable quick coupler. (NFPA 99 2018 5.1.5.1)

Each outlet/inlet shall be legibly identified in accordance with 5.1.11.3. (NFPA 99 2018 5.1.5.5)

Each station outlet/inlet, including those mounted in columns, hose reels, ceiling tracks, or other special installations, shall be designed so that parts or components that are required to be gas specific for compliance with 5.1.5.1 and 5.1.5.9 cannot be interchanged between the station outlet/inlet for different gases. (NFPA 99 2018 5.1.5.7)

All gas outlets with a gauge pressure of 345 kPa (50 psi), including, but not limited to, oxygen, nitrous oxide, medical air, and carbon dioxide, shall deliver 100 SLPM (3.5 SCFM) with a pressure drop of not more than 35 kPa (5 psi) and static pressure of 345 kPa to 380 kPa (50 psi to 55 psi). (NFPA 99 2018 5.1.12.4.10.2)

Support gas outlets shall deliver 140 SLPM (5.0 SCFM) with a pressure drop of not more than 35 kPa (5 psi) gauge and static pressure of 1100 kPa to 1275 kPa (160 psi to 185 psi) gauge. (NFPA 99 2018 5.1.12.4.10.3)

Medical–surgical vacuum inlets shall draw 85 Nl/min (3 SCFM) without reducing the vacuum pressure below 300 mm (12 in.) gauge HgV at any adjacent station inlet. (NFPA 99 2018 5.1.12.4.10.4)

Oxygen and medical air outlets serving critical care areas shall allow a transient flow rate of 170 SLPM (6 SCFM) for 3 seconds. (NFPA 99 2018 5.1.12.4.10.5)

Medical gas concentration & flow rate requirements. (NFPA 99 2018 Table 5.1.12.4.11)

GAS	FLOW RATE	CONCENTRATION
OXYGEN	Non-critical 3.5 SCFM Critical 6.0 SCFM	≥99% oxygen
MEDICAL AIR	Non-critical 3.5 SCFM Critical 6.0 SCFM	19.5%–23.5% oxygen
NITROUS OXIDE	3.5 SCFM	≥99% nitrous oxide
NITROGEN	5.0 SCFM	≤1% O2 or ≥99% N2
VACUUM	3.0 SCFM	N/A
WAGD	3.0 SCFM	N/A
OTHER GASES	3.5 SCFM	As specified by ±1%, unless otherwise specified

MGI INC.
MEDICAL GAS INSTALLERS

OUTLET ANALYSIS

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

OUTLET LOCATION	GAS TYPE & OUTLET QUANTITY				OUTLET SPECS			Leakage w/ Adapter	Leakage w/o Adapter	Operational Flow	Static Pressure	Concentration	Condition & Labeling	PASS	FAIL	PARTICULATE CONTAMINATION	COMMENTS	REPAIRED BY
1ST FLOOR	OXYGEN	MEDICAL AIR	VACUUM	N2														
BIOMED	1	1	1	1	ALLIED	CNT2	CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	N2 IS ALLIED - 500 - DISS	
3RD FLOOR																		
B309	1	1	1		ALLIED	500	CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
B312	1	1	1		ALLIED	500	CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
B315	1	1	1		ALLIED	500	CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
B317	1	1	1		ALLIED	500	CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
TOTAL COUNT:	5	5	5	1														

OUTLET ANALYSIS

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

OUTLET LOCATION	GAS TYPE & OUTLET QUANTITY			OUTLET SPECS	Leakage w/ Adapter	Leakage w/o Adapter	Operational Flow	Static Pressure	Concentration	Condition & Labeling	PASS	FAIL	PARTICULATE CONTAMINATION	COMMENTS	REPAIRED BY
4TH FLOOR	OXYGEN	MEDICAL AIR	VACUUM												
ASU BAY 1	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 2	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 3	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 4	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 5	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 6	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 7	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 8	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 9	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 10	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 11	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 12	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 13	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 14	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 15	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
ASU BAY 16	2	2	3	ALLIED 400 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
B457	1		2	ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	BACK WALL VAC IS ALLIED - CNT2 - CH	
B459	1		2	ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	BACK WALL VAC IS ALLIED - CNT2 - CH	
DENTAL ROOM B403	1		1	ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
DENTAL ROOM B409	1		1	ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
DENTAL ROOM B412	1		1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
DENTAL ROOM B413	1		1	ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
DENTAL ROOM B414	1		1	ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
DENTAL ROOM B415	1		1	ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
DENTAL ROOM B416	1		1	ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
TOTAL COUNT:	41	32	59												

MGI INC. MEDICAL GAS INSTALLERS

OUTLET ANALYSIS

EL PASO VAMC
MGI TECHNICIAN: BRAYDEN SEGER

OUTLET LOCATION	GAS TYPE & OUTLET QUANTITY						OUTLET SPECS	Leakage w/ Adapter	Leakage w/o Adapter	Operational Flow	Static Pressure	Concentration	Condition & Labeling	PASS	FAIL	PARTICULATE CONTAMINATION	COMMENTS	REPAIRED
OR'S	OXY	MED AIR	MED VAC	N2O	N2	WAGD												
OR-1 RIGHT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-1 LEFT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-1 WALL			3		1		ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	N2 IS CHEMETRON - CP - DISS	
OR-2 RIGHT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-2 LEFT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-2 WALL			3		1		ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	N2 IS CHEMETRON - CP - DISS	
OR-3 RIGHT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-3 LEFT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-3 WALL			3		1		ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	N2 IS CHEMETRON - CP - DISS	
OR-4 RIGHT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-4 LEFT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-4 WALL			3		1		ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	N2 IS CHEMETRON - CP - DISS	
OR-5 RIGHT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-5 LEFT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-5 WALL			3		1		ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	N2 IS CHEMETRON - CP - DISS	
OR-6 RIGHT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-6 LEFT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✗	✓	✓	✓	✓		✗	<1MG ALL OUTLETS	N2O LEAKS W/O ADAPTER INSERTED	
OR-6 WALL			3		1		ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	N2 IS CHEMETRON - CP - DISS	
OR-7 RIGHT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-7 LEFT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-7 WALL			3		1		ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	N2 IS CHEMETRON - CP - DISS	
OR-8 RIGHT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-8 LEFT COLUMN	2	1	1	1	1	1	ALLIED CNT2 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS		
OR-8 WALL			3		1		ALLIED 500 CH	✓	✓	✓	✓	✓	✓	✓		<1MG ALL OUTLETS	N2 IS CHEMETRON - CP - DISS	
TOTAL COUNT:	32	16	40	16	24	16												

MEDICAL GAS SOURCE EQUIPMENT ANALYSIS

MGI INC.

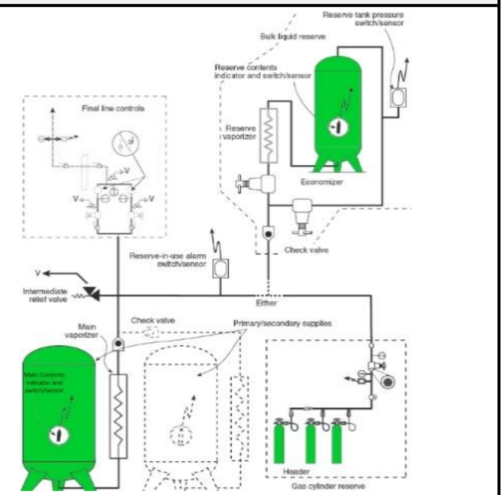
MEDICAL GAS INSTALLERS

MGI™ INC.
MEDICAL GAS INSTALLERS

BULK OXYGEN
SOURCE EVALUATION DATA

LOCATION	BULK PAD
AREA(S) OF FACILITY SERVED	ENTIRE FACILITY
BULK SUPPLIER	AIRGAS
SYSTEM TYPE	LIQ X 18 HP RESERVE
PRIMARY LIQUID LEVEL	53"
PRIMARY PRESSURE	155 PSI
RESERVE LIQUID LEVEL	N/A
RESERVE PRESSURE	2200 PSI
PRIMARY FINAL LINE PRESSURE	55 PSI
RESERVE FINAL LINE PRESSURE	55 PSI
SOURCE VALVE PRESSURE	53 PSI
MAIN VALVE PRESSURE	54 PSI
SOURCE VALVE TYPE	BALL
SOURCE VALVE LOCATION	AT SOURCE
PRESSURE SWITCH LOCATION	C100 MECHANICAL ROOM
EOSC VALVE SIZE	1" BALL
EOSC LOCATION(S)	OUTSIDE DOCK

NFPA 99 2018 SCHEMATIC



NFPA 99 2018 CHECKLIST	PASS	FAIL	N/A
System visibly complies with above schematic per NFPA 99.	✓		
Enclosure (min 6' height) non-combustible, proper ventilation & 2 lockable entries/exits	✓		
Containers permanently anchored & cylinders adequately secured	✓		
Noncombustible pad w/3' maintenance clearance, accessible for delivery & pad adequately sized for vehicles	✓		
Top & bottom fill, hose purge, & vent valves present	✓		
Proper fill circuit (inlet, strainer, check valve, purge valve, supports)	✓		
Piped w/brazed copper, brass, or stainless & no flex connects/fittings wear & leaks	✓		
Reserve and primary check valves in place for reserve to feed automatically	✓		
Reserve economizer in place to feed upstream final regulators	✓		
Drainage prevented towards (buildings, drains, etc.) & from hazards	✓		
Labeling: pipeline, source/main valves & "Oxygen-No Smoking, No Open Flames"	✓		
Vaporizer appears adequately sized & isolation valves arranged for service w/o supply interruption	✓		
Emergency Supply (3' clearance) or (IBER) w/2 checks, secured, & labeled	✓		
Distance from the following: 1' - all structures (Type I & II), * 50' - wood framed Type III, IV, or V building types, * 50' - Public assembly or nearest non- ambulatory patient, * 5' - Property lines & overhead electrical wiring, 10' building openings, * 10' - public sidewalks & parked vehicles, * 15' - Hazardous piping materials & below ground flammable liquid tanks, * 25' - 0-1000 gal flammable liquids/liquefied gas, below ground flammable liquid fill or vent, 0-25,000 scf flammable gases, or slow burning solids, * 50' - >1000 gal flammable liquids/liquefied gas, >25,000 scf flammable gases, or rapidly burning solids, 3' - combustible surfaces (asphalt & expansion joint filler) where liquid oxygen may fall, 8' - Sewer or drain openings to delivery connects, reliefs, mobile supply equipment, & liquid withdrawal connects, Walls may not form a 3-4 sided court.	✓		
MEETS ALL NFPA 99 2018 CODE REQUIREMENTS		PASS	

ADDITIONAL COMMENTS

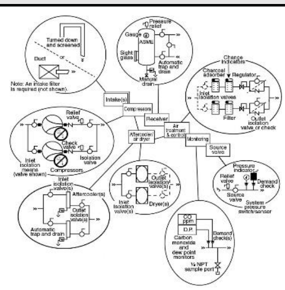
MGI INC. MEDICAL GAS INSTALLERS

MEDICAL AIR SYSTEM
SOURCE EVALUATION DATA

LOCATION	C106 MECHANICAL ROOM	**LINE PRESSURE**	50 PSI
AREA(S) OF FACILITY SERVED	ENTIRE FACILITY	**DRYER TYPE**	DESICCANT
MANUFACTURER	POWEREX	**DRYER INFO**	PMD35000AJ
MODEL NUMBER	MPD10084P5	**DEW POINT MONITOR INFO**	POWEREX BUILT IN
SERIAL NUMBER	(H)2/26/2019-000001539-425	**CO MONITOR INFO**	ENMET CO GUARD
SYSTEM TYPE	RECIPROCATING	**SOURCE VALVE LOCATION**	AT SOURCE
CONFIGURATION	DUPLEX	**PRESSURE SWITCH LOCATION**	AT SOURCE
PUMP MAKE AND MODEL	N/A	**PUMP 1 HOURS**	3172
HORSEPOWER	10 H.P.	**PUMP 2 HOURS**	3571
LEAD ON/OFF	95 PSI / 115 PSI	**PUMP 3 HOURS**	N/A
LAG ON/OFF	65 PSI / 85 PSI	**PUMP 4 HOURS**	N/A

MEDICAL AIR COMP MAINTENANCE REQUIRED

NFPA 99 2018 SCHEMATIC



MEDICAL AIR PURITY ANALYSIS

ELEMENT	NFPA LIMIT	READING	PASS/FAIL
DEW POINT (MONITOR ON SYSTEM)	35 DEG F	-75 DEG F	PASS
DEW POINT (CS MONITOR)	35 DEG F	-55 DEG F	PASS
CARBON MONOXIDE (MONITOR ON SYSTEM)	10 PPM	0 PPM	PASS
CARBON MONOXIDE (CS MONITOR)	10 PPM	0 PPM	PASS
CARBON DIOXIDE	500 PPM	97 PPM	PASS
METHANE	25 PPM	1 PPM	PASS
CHLOROFORM	2 PPM	0 PPM	PASS
OXYGEN	19.5%-23.5%	21.3%	PASS
COMMENTS			

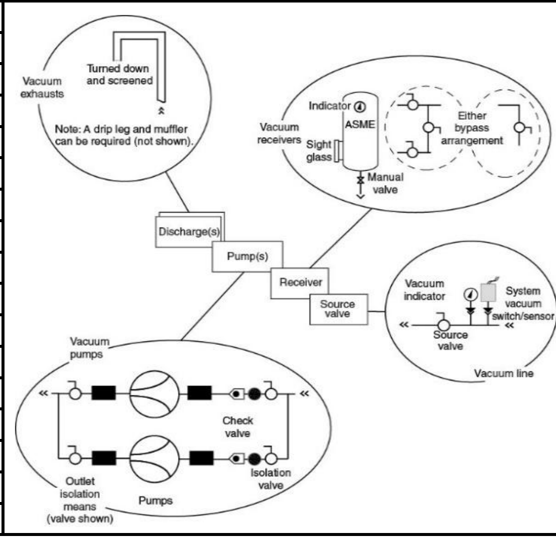
NFPA 99 2018 CHECKLIST	PASS	FAIL	N/A
System visibly complies with above schematic per NFPA 99.	✓		
Indoors in a dedicated mechanical equipment area, adequately ventilated and with any required utilities	✓		
System has two or more medical air compressors	✓		
System is supplied with flexible connectors on inlet and outlet sides of the compressors	✓		
System power supplied from emergency power	✓		
System is equipped with manual disconnects	✓		
System is equipped with a local lag, dew point and CO alarm	✓		
The local alarms are wired to the master alarm panel	✓		
Receiver tank is supplied with an Auto drain, pressure gauge, sight glass, pressure relief, & ASME label	✓		
There is a receiver bypass valve so the receiver can be serviced without interruption of the system	✓		
Each pump has an isolation valve so a pump can be serviced without interruption of the system	✓		
System automatically alternates the pumps evenly	✓		
Duplexed dryer system	✓		
1/4" sample port installed	✓		
Source valve in proper location and labeled	✓		
Mainline and intake piping properly labeled	✓		
Intake above roof level, >20' above ground, 10 ft from doors, windows, other openings, and 25' from exhausts	✓		
Intake turned down and screened	✓		
MEETS ALL NFPA 99 2018 CODE REQUIREMENTS	**PASS**		

ADDITIONAL COMMENTS

MEDICAL VACUUM SYSTEM
SOURCE EVALUATION DATA

LOCATION	C106 MECHANICAL ROOM
AREA(S) OF FACILITY SERVED	ENTIRE FACILITY
MANUFACTURER	DEKKER
MODEL NUMBER	VMX0203KA3-40-ES
SERIAL NUMBER	119576
SYSTEM TYPE	LIQUID RING
CONFIGURATION	TRIPLEX
PUMP MAKE AND MODEL	DEKKER DV0200B-KA3
HORSEPOWER	15 H.P.
LEAD ON/OFF	19" HG / 24" HG
LAG ON/OFF	14" HG / 19" HG
LINE PRESSURE	24" HG
SOURCE VALVE LOCATION	AT SOURCE
PRESSURE SWITCH LOCATION	AT SOURCE
PUMP 1 HOURS	37630
PUMP 2 HOURS	41941
PUMP 3 HOURS	31451
PUMP 4 HOURS	N/A

NFPA 99 2018 SCHEMATIC



NFPA 99 2018 CHECKLIST	PASS	FAIL	N/A
System visibly complies with above schematic per NFPA 99.	✓		
Located in a dedicated mechanical equipment area, adequately ventilated and with any required utilities	✓		
System has two or more medical vacuum pumps	✓		
System has flexible connectors	✓		
System has antivibration mountings	✓		
System power supplied from emergency power	✓		
System is equipped with manual disconnects	✓		
System is equipped with a local lag alarm	✓		
The local alarms are wired to the master alarm panel	✓		
Receiver tank is supplied with an Auto drain & ASME label	✓		
There is a receiver bypass valve so the receiver can be serviced without interruption of the system	✓		
Each pump has an isolation valve so a pump can be serviced without interruption of the system	✓		
System automatically alternates the pumps evenly	✓		
Source valve in proper location and labeled	✓		
Mainline and exhaust piping properly labeled	✓		
Exhaust is turned down and screened	✓		
Exhaust ≥25' from doors, windows, intakes, building openings & public assembly & copper or stainless, no dips, drip legs installed, & sufficient diameter	✓		
MEETS ALL NFPA 99 2018 CODE REQUIREMENTS	PASS		

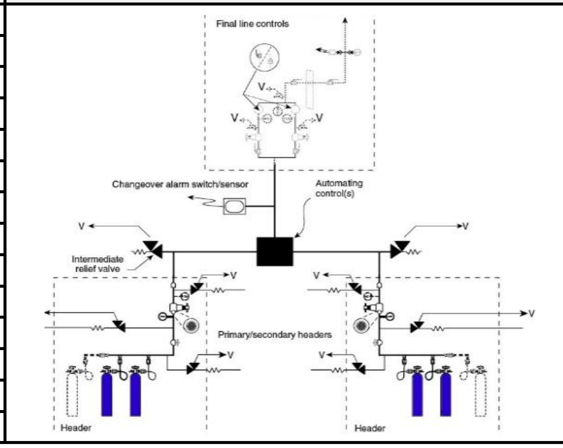
ADDITIONAL COMMENTS

MGI INC.
MEDICAL GAS INSTALLERS

NITROUS OXIDE MANIFOLD SYSTEM SOURCE EVALUATION DATA

LOCATION	NITROUS OXIDE MANIFOLD ROOM
AREA(S) OF FACILITY SERVED	OR'S
MANUFACTURER	CHEMETRON
MODEL NUMBER	4000
SERIAL NUMBER	20111206003
SYSTEM TYPE	HP X HP
CYLINDERS ON RIGHT BANK	6
CYLINDERS ON LEFT BANK	6
RIGHT BANK PRESSURE	600 PSI
LEFT BANK PRESSURE	600 PSI
CHANGE OVER PRESSURE	150 PSI
LINE PRESSURE	55 PSI
SOURCE VALVE TYPE	BALL
SOURCE VALVE LOCATION	AT SOURCE
PRESSURE SWITCH LOCATION	C100 MECHANICAL ROOM

NFPA 99 2018 SCHEMATIC



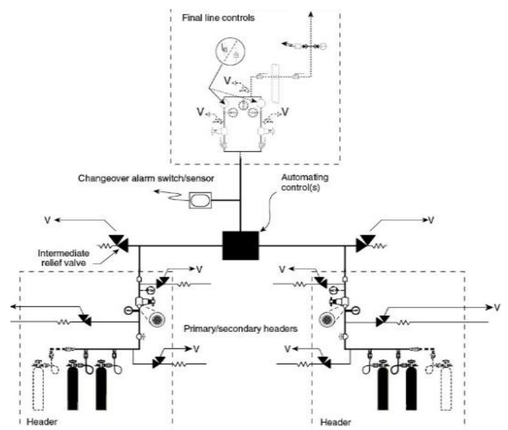
NFPA 99 2018 CHECKLIST	PASS	FAIL	N/A
System visibly complies with above schematic per NFPA 99.	✓		
Secured with lockable doors or gates or otherwise secured.	✓		
1 hour fire rated door and walls.	✓		
Source valve labeled per NFPA 99.	✓		
Proper natural ventilation	✓		
Proper mechanical ventilation			N/A
Manifold power supplied from emergency power	✓		
Automatic bank alternation	✓		
Warning lights when banks alternate	✓		
Relief Valves vented outside, pipe size, turned down, & screened	✓		
No flammable materials, cylinders containing flammable gases, or containers containing flammable liquids	✓		
Electrical devices protected in room	✓		
Cylinders secured with racks or chains	✓		
Headers have header valve and check valves	✓		
Cylinders prevented from reaching temperatures in excess 130°F. and lower than 20°F.	✓		
Door Labeling: CAUTION, Medical Gases, No Smoking or Open Flame, Room May Have Insufficient Oxygen Open Door & Allow Room to Ventilate before Entering	✓		
MEETS ALL NFPA 99 2018 CODE REQUIREMENTS	PASS		

ADDITIONAL COMMENTS

NITROGEN MANIFOLD SYSTEM
SOURCE EVALUATION DATA

LOCATION	NITROGEN MANIFOLD ROOM
AREA(S) OF FACILITY SERVED	BIOMED AND OR'S
MANUFACTURER	CHEMETRON
MODEL NUMBER	4000
SERIAL NUMBER	20111209001
SYSTEM TYPE	HP X HP
CYLINDERS ON RIGHT BANK	6
CYLINDERS ON LEFT BANK	6
RIGHT BANK PRESSURE	2000 PSI
LEFT BANK PRESSURE	1400 PSI
CHANGE OVER PRESSURE	250 PSI
LINE PRESSURE	165 PSI
SOURCE VALVE TYPE	BALL
SOURCE VALVE LOCATION	AT SOURCE
PRESSURE SWITCH LOCATION	C100 MECHANICAL ROOM

NFPA 99 2018 SCHEMATIC



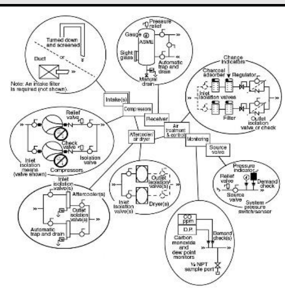
NFPA 99 2018 CHECKLIST	**PASS**	**FAIL**	**N/A**
System visibly complies with above schematic per NFPA 99.	✓		
Secured with lockable doors or gates or otherwise secured.	✓		
1 hour fire rated door and walls.	✓		
Source valve labeled per NFPA 99.	✓		
Proper natural ventilation	✓		
Proper mechanical ventilation			**N/A**
Manifold power supplied from emergency power	✓		
Automatic bank alternation	✓		
Warning lights when banks alternate	✓		
Relief Valves vented outside, pipe size, turned down, & screened	✓		
No flammable materials, cylinders containing flammable gases, or containers containing flammable liquids	✓		
Electrical devices protected in room	✓		
Cylinders secured with racks or chains	✓		
Headers have header valve and check valves	✓		
Cylinders prevented from reaching temperatures in excess 130°F. and lower than 20°F.	✓		
Door Labeling: CAUTION, Medical Gases, No Smoking or Open Flame, Room May Have Insufficient Oxygen Open Door & Allow Room to Ventilate before Entering	✓		
MEETS ALL NFPA 99 2018 CODE REQUIREMENTS		**PASS**	

ADDITIONAL COMMENTS

DENTAL AIR SYSTEM
SOURCE EVALUATION DATA

LOCATION	A421 MECHANICAL ROOM	**LINE PRESSURE**	50 PSI
AREA(S) OF FACILITY SERVED	DENTAL CLINIC	**DRYER TYPE**	REFRIGERANT
MANUFACTURER	POWEREX	**DRYER INFO**	HANKISON HPR50
MODEL NUMBER	SEQ200740AJ	**DEW POINT MONITOR INFO**	NEWPORT SCIENTIFICS 1092E
SERIAL NUMBER	(M)3/18/2019-S1539-905	**CO MONITOR INFO**	DYNAMATION ABL-4021
SYSTEM TYPE	ENCLOSED SCROLL	**SOURCE VALVE LOCATION**	AT SOURCE
CONFIGURATION	QUADPLEX	**PRESSURE SWITCH LOCATION**	AT SOURCE
PUMP MAKE AND MODEL	POWEREX	**PUMP 1 HOURS**	1597
HORSEPOWER	20 H.P.	**PUMP 2 HOURS**	1648
LEAD ON/OFF	95 PSI / 115 PSI	**PUMP 3 HOURS**	1651
LAG ON/OFF	65 PSI / 85 PSI	**PUMP 4 HOURS**	1642

NFPA 99 SCHEMATIC



DENTAL AIR PURITY ANALYSIS

ELEMENT	NFPA LIMIT	READING	PASS/FAIL
DEW POINT (MONITOR ON SYSTEM)	35 DEG F	53 DEG F	FAIL
DEW POINT (CS MONITOR)	35 DEG F	50 DEG F	FAIL
CARBON MONOXIDE (MONITOR ON SYSTEM)	10 PPM	0 PPM	PASS
CARBON MONOXIDE (CS MONITOR)	10 PPM	0 PPM	PASS
COMMENTS			

NFPA 99 CHECKLIST	PASS	FAIL	N/A
System visibly complies with above schematic per NFPA 99.	✓		
Indoors in a dedicated mechanical equipment area, adequately ventilated and with any required utilities	✓		
System has two or more instrument air compressors or a standby header	✓		
System is supplied with flexible connectors on inlet and outlet sides of the compressors	✓		
System power supplied from emergency power	✓		
System is equipped with manual disconnects	✓		
System is equipped with a local lag and dew point alarm	✓		
The local alarms are wired to the master alarm panel	✓		
Receiver tank is supplied with an Auto drain, pressure gauge, sight glass, pressure relief, & ASME label	✓		
There is a receiver bypass valve so the receiver can be serviced without interruption of the system	✓		
Each pump has an isolation valve so a pump can be serviced without interruption of the system	✓		
System automatically alternates the pumps evenly	✓		
Duplexed dryer system	✓		
1/4" sample port installed	✓		
Source valve in proper location and labeled	✓		
Mainline and intake piping properly labeled	✓		
MEETS ALL NFPA 99 CODE REQUIREMENTS		PASS	

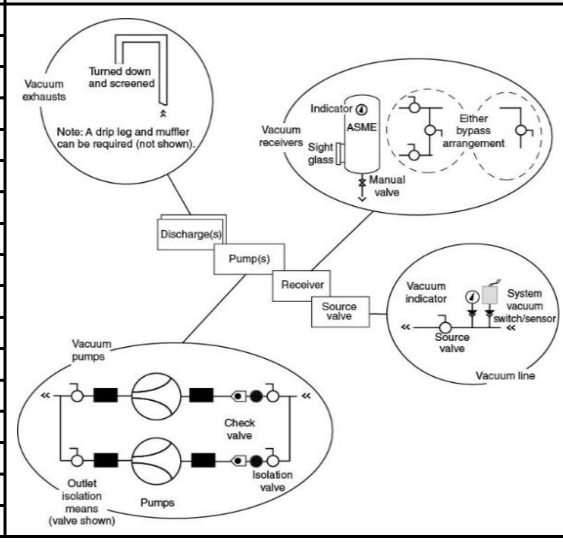
ADDITIONAL COMMENTS

FACILITY IS AWARE OF THE HIGH DEWPOINT. THEY HAVE ORDERED A NEW DRYER TO FIX THIS ISSUE.

MEDICAL VACUUM SYSTEM
SOURCE EVALUATION DATA

LOCATION	A421 MECHANICAL ROOM
AREA(S) OF FACILITY SERVED	DENTAL CLINIC
MANUFACTURER	TECH WEST
MODEL NUMBER	VPD30D5
SERIAL NUMBER	VPD30-5038
SYSTEM TYPE	TURBINE
CONFIGURATION	DUPLEX
PUMP MAKE AND MODEL	N/A
HORSEPOWER	7.5 H.P.
LEAD ON/OFF	7" HG / 10" HG
LAG ON/OFF	5" HG / 7" HG
LINE PRESSURE	10" HG
SOURCE VALVE LOCATION	AT SOURCE
PRESSURE SWITCH LOCATION	N/A
PUMP 1 HOURS	10366
PUMP 2 HOURS	10193
PUMP 3 HOURS	N/A
PUMP 4 HOURS	N/A

NFPA 99 2018 SCHEMATIC



NFPA 99 2018 CHECKLIST	PASS	FAIL	N/A
System visibly complies with above schematic per NFPA 99.	✓		
Located in a dedicated mechanical equipment area, adequately ventilated and with any required utilities	✓		
System has two or more medical vacuum pumps	✓		
System has flexible connectors	✓		
System has antivibration mountings	✓		
System power supplied from emergency power	✓		
System is equipped with manual disconnects	✓		
System is equipped with a local lag alarm			N/A
The local alarms are wired to the master alarm panel			N/A
Receiver tank is supplied with an Auto drain & ASME label	✓		
There is a receiver bypass valve so the receiver can be serviced without interruption of the system	✓		
Each pump has an isolation valve so a pump can be serviced without interruption of the system	✓		
System automatically alternates the pumps evenly	✓		
Source valve in proper location and labeled	✓		
Mainline and exhaust piping properly labeled	✓		
Exhaust is turned down and screened			N/A
Exhaust ≥25' from doors, windows, intakes, building openings & public assembly & copper or stainless, no dips, drip legs installed, & sufficient diameter			N/A
MEETS ALL NFPA 99 2018 CODE REQUIREMENTS	PASS		

ADDITIONAL COMMENTS

END OF REPORT

MEDICAL GAS INSTALLERS

EL PASO VAMC
5001 N PIEDRAS ST
EL PASO, TX 79930